

Evan Frangipane

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Education

Ph.D. in Physics *University of California, Santa Cruz*

Santa Cruz, CA *Aug 2018 - Aug 2024*

B.A. in Physics *University of California, Berkeley*

Berkeley, CA *Aug 2014 - Jun 2018*

Employment

Graduate Research Scientist, *University of California, Santa Cruz*

Aug 2018 - Aug 2024

- Ran large-scale microlensing simulations on a distributed Kubernetes cluster and applied Anderson-Darling hypothesis testing to [distinguish planet vs. black hole populations](#).

Graduate Student Instructor, *University of California, Santa Cruz*

Aug 2018 - Aug 2024

- Taught data analysis on real physics datasets, including hypothesis testing with Chi-square tests, count data using Poisson distributions, and linear regression and machine learning techniques to test theoretical models.

Quantitative Research Assistant, *Lawrence Berkeley National Laboratory & INFN Genoa*

Sept 2016 - Jun 2018

- Applied Monte Carlo particle track generation to model detector responses, analyzed material interactions to establish hit thresholds, processed and structured simulation outputs, and created visualizations to communicate results.

Quantitative Research Intern, *NASA Goddard Space Flight Center*

Jun 2016 - Aug 2016

- Forecasted solar flares using NASA solar datasets, applied the Heidke Skill Score to identify optimal prediction thresholds, and won NASA Intern Award in 2016.

Technical Projects

Streaming Anomaly Detection Dashboard (Self-directed)

August 2025

- Built an [interactive Streamlit dashboard](#) to flag suspicious activity in synthetic music streaming data.
- Applied time series methods, signal processing, and Isolation Forest to detect anomalous user behavior patterns.

GenAI and RAG Tools for Physics Research (Self-Directed)

July 2025

- Fine-tuned TinyLlama with LoRA using Hugging Face Transformers to improve [physics abstract generation](#).
- Built a [retrieval-augmented chatbot \("arxbot"\)](#) by combining BM25 retrieval with local LLM inference via Ollama.

End-to-End arXiv Abstract Classification (Self-Directed)

July 2025

- Developed and deployed a logistic regression model for [real-time classification of arXiv abstracts](#), enabling live prediction serving in production.
- Experimented with DistilBERT fine-tuning (via LoRA), but found training costs and marginal gains did not justify deployment over logistic regression.

Multi-Model Fraud Detection: LR, SVM, and Random Forest (Self-directed)

Feb 2025

- Compared three models for [credit card fraud detection](#) on highly imbalanced (280k transactions, < 0.2% fraud) dataset.
- Random Forest outperformed logistic regression and SVM with an 0.84 AUCPR and 0.82 f1-score.

Skills

- **Programming & Tools:** Python (Pandas, scikit-learn, Pytorch), SQL, Git, Tableau, $\text{\LaTeX}$
- **Machine Learning & Statistics:** Supervised & Unsupervised Learning, Regression, Hypothesis Testing, Time Series, Natural Language Processing, LLMs & Retrieval-Augmented Generation (RAG)
- **MLOps:** Model Deployment (FastAPI, Render, AWS Lambda), Containerization (Docker), Orchestration (Kubernetes), Experiment Tracking and Reproducibility (MLFlow)
- **Professional Skills:** Data-driven decision-making, stakeholder engagement, and impactful data storytelling